

Design and Analysis of a 1-bit Comparator for Low Power VLSI Systems

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Abstract— In contemporary VLSI systems, the requirement for high-speed, low-power digital circuits has made the design of effective arithmetic and logic units crucial. Processors and communication systems depend on comparators, which need to be optimized for space, speed, and power. Using a hybrid technique that combines Transmission Gate Logic (TGL), Gate Diffusion Input (GDI), and Pass Transistor Logic (PTL), this paper suggests a one-bit comparator. To reduce parasitic capacitance and transistor count, the design uses a 6T-MUX-based full adder. It also uses a TGL-based AND gate to increase output speed and a GDI structure to improve logic efficiency while using less power. The circuit is simulated with Spectre and constructed in Cadence Virtuoso with the GPDK 90nm technology node. Without sacrificing size or performance, the results provide a 58.7% power reduction and a 31.2% delay improvement over traditional CMOS comparators. The design is now appropriate for low-power applications of the future. Future development will involve adopting advanced technology nodes and expanding to multi-bit comparators.

Keywords- One-Bit Comparator, Transmission Gate Logic (TGL), Low Power Design, MUX-6T Adder, NAND Gate, CMOS optimization.

I. INTRODUCTION

The design and implementation of a one-bit comparator employing a MUX-6T full adder based on Gate Diffusion Input (GDI) and Pass Transistor Logic (PTL) is shown in this work. A vital part of digital systems, the comparator finds extensive use in microprocessors, signal processing applications, and arithmetic and logic units (ALUs). The Boolean expressions for a 1-bit comparator are given as:

$$A > B = A \cdot B' \quad (1)$$

$$A < B = A' \cdot B \quad (2)$$

$$A == B = A \odot B \quad (3)$$

Equations (1) to (3) demonstrate the three basic Boolean criteria that underpin the operation of the suggested one-bit comparator. Each of these formulas outlines the reasoning for identifying "greater than," "less than," and "equal to" conditions between two binary inputs, A and B. Basic logic gates are mapped to these logical expressions in the comparator architecture depicted in Figure 1. Inverted A and direct B inputs are used to drive an AND gate, which implements the "less than" condition (Eq. 2). The "greater than" condition (Eq. 1) is achieved by means of a specific

adder block that only produces a high output when A surpasses B. Because the XNOR gate naturally produces a high output when both inputs are identical, it is utilized in the "equal to" condition (Eq. 3). Inverters are used to extract complemented signals when needed, which is consistent with the logic specified in the equations. This modular logic-based approach guarantees low power consumption and reduced delay, making the design effective and appropriate for integration in low-power digital systems.

The solution uses several cutting-edge logic approaches, such as Transmission Gate Logic (TGL) and MUX-based complete adders. The GDI technique is used to further increase the full adder's efficiency, while the MUX-6T PTL approach is used to lower the number of transistors and power consumption [1]. A crucial part of the comparator, the full adder, is made with a multiplexer-based architecture that provides a small and effective way to do arithmetic operations. This method is appropriate for low-power applications since it minimizes both the circuit's space and complexity [2].

The literature has examined several kinds of adders, each with unique benefits and drawbacks, including ripple-carry adders, carry-lookahead adders, and carry-select adders [3]. Because fewer transistors are needed, the MUX-based adder operates more quickly and with less power. Furthermore, by lowering the latency and power consumption, the adder design's usage of the GDI approach improves performance even further, making it the perfect option for high-speed and low-power applications [4].

The output AND gate's usage of Transmission Gate Logic (TGL) is one of the implementation's main features. Compared to conventional CMOS logic, TGL has several benefits, such as decreased power consumption, enhanced noise immunity, and decreased propagation time [5]. To design a comparator more quickly and efficiently by using TGL, which is essential for high-performance digital systems. This design's use of TGL and MUX-based full adders shows how sophisticated logic approaches can be used to create efficient digital circuits.

The aim of this work is to develop and evaluate a one-bit comparator that achieves an optimal trade-off between power usage, speed, and area for cutting-edge VLSI applications. By combining three effective logic design methods—Pass

Transistor Logic (PTL), Gate Diffusion Input (GDI), and Transmission Gate Logic (TGL)—the aim is to establish a hybrid architecture that addresses the shortcomings of traditional CMOS-based comparators. The main emphasis is on minimizing overall power consumption while improving switching performance, all without sacrificing logical functionality or increasing the number of transistors. This strategy is intended for low-power and high-speed operation, making the comparator ideal for integration into contemporary digital systems, such as embedded processors and communication circuits, where energy efficiency and performance are essential. The design is carried out within the Cadence Virtuoso framework utilizing the GPD90nm technology node, and simulations are performed with the Spectre simulator to assess the power, speed, and area characteristics of the comparator.

The paper is structured in detail as follows: Section I is the Introduction, Section II consists of the Literature Review, Section III discusses the Methodology and Implementation, Section IV offers the Results and Discussion, and Section V provides the conclusion of the work.

II. LITERATURE REVIEW

Recent investigations have highlighted the importance of developing digital circuits that exhibit both rapid operation and low power usage. An energy-efficient digital architecture was introduced in [1] featuring a novel 6-T multiplexer-based full adder, specifically optimized to minimize power consumption and reduce leakage current. To enhance performance and efficiency, a low-power, high-speed comparator was created utilizing a multiplexer-based complete adder cell, targeting mobile communication systems [2]. An area-efficient design for a 1-bit comparator was presented, leveraging a hybrid full adder module that integrates Pass Transistor Logic (PTL) and Gate-Diffusion Input (GDI) logic in order to emphasize area and power savings in digital circuits [3].

A fundamental aspect of low-power VLSI systems is the Gate-Diffusion Input (GDI) technique, initially described in [4] as a groundbreaking approach for generating power-efficient digital combinatorial circuits. To accomplish low power usage coupled with high-speed performance, a cell design methodology utilizing transmission gates was proposed to create balanced XOR-XNOR circuits in hybrid-CMOS logic [5]. These studies illustrate how essential logic optimization techniques are for achieving significant power reductions and improved performance in modern digital systems.

Recent advancements in digital circuit design have concentrated on attaining both area and power efficiency, particularly for applications related to frequency division and key generation. A novel diffused bit generator that is area and power-efficient, based on AVLS-TSPC, was suggested and fine-tuned for key generation, demonstrating significant improvements in energy efficiency and circuit performance [6]. This work was further developed by introducing a low-power 16/17 dual modulus prescaler design utilizing TSPC-AVLS technology to enhance operational speed while reducing power consumption in communication systems [7]. Additionally, a 32/33 pre-scaler for frequency dividers grounded in AVLS technology was proposed, focusing on energy efficiency and compact design for high-performance electronic applications [8].

Numerous studies have also been conducted on the use of entire adder modules in comparator design. Complete adder modules have proven to be effective and versatile in digital circuit applications when used in comparator design [9]. Furthermore, a 1-bit full adder module was used to construct a 4-bit comparator with a focus on area and power efficiency, two aspects that are critical for small and energy-efficient devices [10]. These research support low-power and high-performance computing by highlighting the significance of full adder-based designs in efficient digital circuit and comparator implementations.

There has also been interest in low-power adiabatic logic designs. Low-power square root carry chooses adders and Wallace tree multipliers based on adiabatic logic demonstrated significant power savings and increased efficiency, especially for low-energy applications [11]. The invention of a low-power, high-speed 16-bit square root carry pick adder employing adiabatic logic demonstrated its potential for high-performance applications where power efficiency is critical [12].

Additional advancements include a low-power square root carry select adder based on D flip-flops based on AVLS-TSPC. For energy-efficient arithmetic circuits, this adder effectively reduces power consumption while maintaining calculation speed [13]. Hybrid logic techniques have also been shown to increase digital circuit performance while consuming fewer transistors and electricity. A low-power 8-bit magnitude comparator using a hybrid PTL/CMOS logic approach achieved significant transistor count reduction without compromising operational precision [14]. Additionally, the energy efficiency and scalability of PTL-based full adders for low-power applications were confirmed by a comprehensive analysis conducted at multiple nanoscale technology nodes [15]. Together, these works demonstrate the growing significance of hybrid logic designs for tiny and energy-efficient architectures in next-generation integrated circuits. RTL optimization techniques have been effectively employed to enhance the performance of a 32-bit carry select adder, as demonstrated in [16]. This approach improves computation speed and area efficiency, contributing to the design of high-performance, low-power arithmetic units.

Although a lot of work has gone into creating high-speed, energy-efficient arithmetic circuits, especially with the development of full adder architectures employing methods like 6T-MUX-based design, PTL, GDI, TGL, and hybrid logic, the use of these optimized adders in comparator design is still largely unexplored. Full adder cells have been used in previous works for comparator duties [9, 10], although many designs only concentrate on lowering power or area, frequently at the expense of delay or logic complexity. Furthermore, few studies fully utilize the combined advantages of PTL, GDI, and TGL within a single comparator architecture, despite the fact that hybrid logic approaches have demonstrated promise in lowering transistor count and power consumption [14, 15]. Additionally, low-power arithmetic units have been made possible by adiabatic logic and RTL-level optimizations [11, 16], but there is still a lack of integration of these technologies into small, high-performance comparator circuits that are appropriate for scaled-down technologies.

III. METHODOLOGY AND IMPLEMENTATION

The comparator circuit along with the MUX-6T full adder is a circuit that uses the Pass Transistor logic (PTL), to detect

the signals that are coming as input to the adder and the behavior of the adder is utilized to be able to get favorable Boolean expression at sum and carry of the adder, which is then used by CMOS logic based circuits that as a result form the main output ports of the comparator circuit.

The design determines if A is greater than, less than, or equal to B by combining NOT and AND gates. To enable bitwise comparison, the complete adder is used with a grounded carry-in, and the XOR operation is crucial for identifying equality. Concurrently, conditions for $A < B$ and $A > B$ are derived using negated inputs, guaranteeing a simple yet efficient implementation. A block diagram of the comparator circuit using a full adder is shown in Figure 1. This block diagram uses an adder and logic gates to depict a 1-bit comparator. Three conditions are determined by the circuit. $A > B$, $A < B$, and $A = B$. [2]

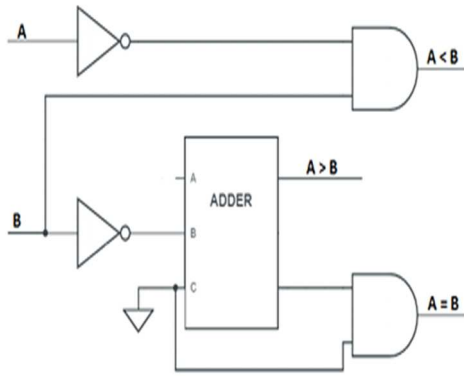


Fig. 1. Block diagram of the 1-bit comparator

A. The Full Adder

A full adder is an essential digital circuit employed in arithmetic computations, especially in the context of binary addition. It accepts three inputs: two binary digits (A and B) and a carry-in (C_{in}), and it generates two outputs: the sum (S) and the carry-out (C_{out}). The sum reflects the outcome of the addition, whereas the carry-out signifies the occurrence of an overflow, which is vital for performing multi-bit additions in more complex systems such as ripple-carry adders. The operation of a full adder is characterized by specific logic equations.

$$SUM = A \oplus B \oplus C \quad (4)$$

$$CARRY = AB + BC + CA \quad (5)$$

Processors, ALUs (Arithmetic Logic Units), and other digital systems that execute binary arithmetic frequently use full adders. Numerous complete adder designs have been put forth over time, each of which optimizes for particular factors, including speed, area, and power dissipation. Because of its small transistor count and effective operation, the MUX-6T full adder is the one chosen in this study to create a 1-bit comparator circuit with less space and power usage.

The MUX-6T complete adder is a small, incredibly effective device that performs binary addition using inverters and Pass Transistor Logic (PTL). Because it uses less power and has fewer transistors than conventional CMOS-based complete adders, this design is especially useful in low-power and space-constrained applications (Ramana Murthy *et al.*, 2012). The circuit uses PTL-implemented multiplexers to

choose between various logic levels according to the input signals. The total number of transistors used in this adder is 14, including the transistors used by the inverters. A block diagram of the MUX-6T adder is shown in Figure 2 [1].

In comparison to conventional adder designs, the multiplexer-based method streamlines the circuit topology, allowing for faster signal propagation and better performance. Furthermore, as leakage power is a concern in deep submicron technologies, the decreased transistor count directly results in lower leakage current. All things considered, the 6-T multiplexer-based full adder provides the best possible balance between speed, scalability, and power efficiency, making it a useful addition to contemporary energy-efficient digital systems. The implementation of the MUX-6T full adder is shown in Figure 3 [1].

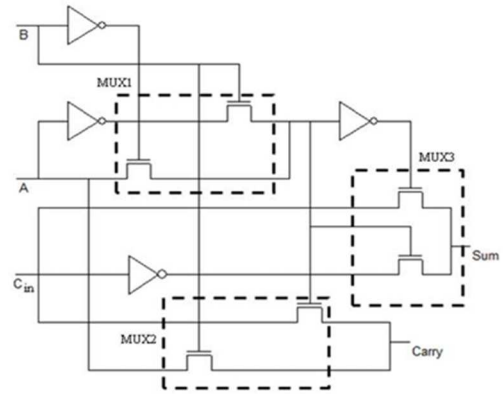


Fig. 2. Block diagram of MUX-6T Full Adder.

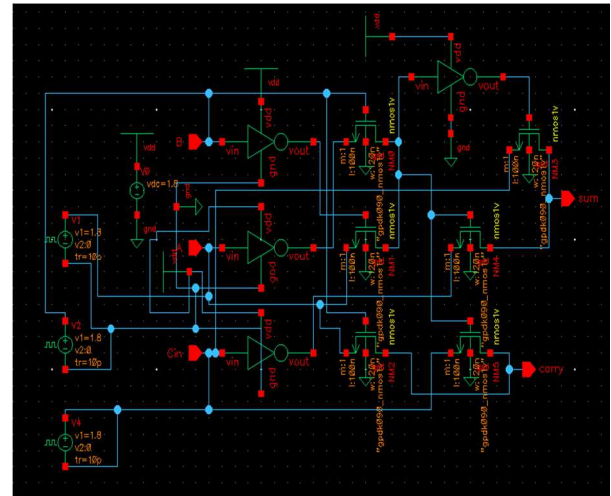


Fig. 3. MUX-6T Full Adder using Pass Transistor logic (PTL).

B. Proposed Comparator

Using a pass-transistor logic (PTL)-based complete adder to increase a comparator's efficiency is beneficial because of its better speed, lower power consumption, and fewer transistors. PTL reduces switching activity and directs pathways to ground, resulting in minimal delay and effective logic implementation. To further optimize the circuit, transmission gate logic (TGL) is used in place of traditional CMOS logic AND and NOT gates at the $A < B$ and $A = B$ outputs.

A direct, low-resistance signal channel is provided by TGL, which consists of a PMOS-NMOS pair. This improves signal integrity and lowers propagation delay. Transmission gates reduce signal deterioration by maintaining full voltage

swing, in contrast to typical pass transistors. This combination of a PTL-based adder and TGL-based output selection greatly increases speed, lowers power consumption, and minimizes the total number of transistors.

A PTL-based complete adder comparator with an optimized design for high-speed and low-power applications is implemented by the suggested circuit. While the B input to the adder is supplied as inverted $B(\bar{B})$, guaranteeing an effective XOR-based computation, the carry-in (C_{in}) is grounded, simplifying the arithmetic operation. With B as the input, a transmission gate (TGL) is employed at the output stage for the $B > A$ condition, in which A controls the PMOS gate and $\bar{A}$ controls the NMOS gate. With this setup, the transmission gate produces $\bar{A}B$, correctly indicating the condition for $A < B$. In order to balance mobility differences and maximize driving strength, the NMOS transistors are sized three times larger than the PMOS transistors. The circuit is ideal for low-power VLSI implementations because it runs at a supply voltage of 1.8V. To ensure that the sum signal reaches the output for equality detection, the same transmission gate (TGL) is utilized at the adder's sum output for $A = B$. The input to the transmission gate is the sum signal. The total number of transistors in this comparator circuit now amounts to 22, which is less than the generic comparator circuits. The implementation of the proposed comparator circuit is shown in Figure 4.

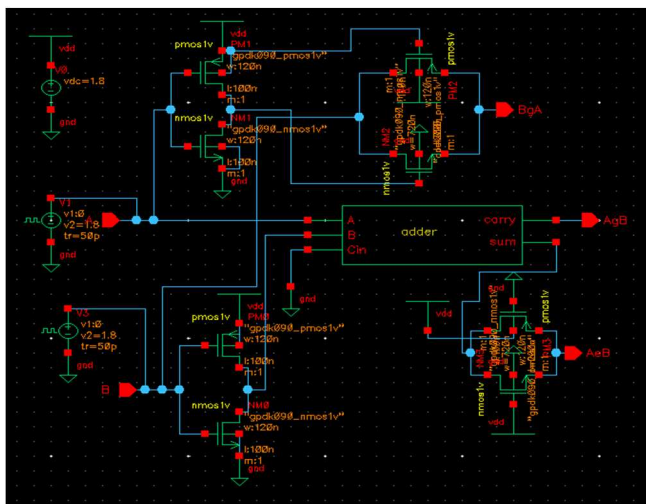


Fig. 4. Proposed 1-bit comparator circuit using transmission gate (TGL)

IV. RESULTS AND DISCUSSION

This work was carried out with 90nm CMOS technology and the Cadence Virtuoso design environment. Within the Virtuoso platform, schematic was created and validated. Using Cadence Spectre simulations, functional verification and performance analysis were conducted. The circuit operated correctly, as evidenced by the successful generation of the output waveform. The generated waveform validated the expected behavior, confirming timing accuracy and design integrity. This implementation reflects industry-standard practices in modern VLSI design and verification workflows.

A. Area

The suggested comparator circuit efficiently maximizes transistor utilization by using **only 22 transistors as compared to a CMOS logic-based implementation which uses 30 transistors**. Because there are fewer transistors, the design is more effective for low-power VLSI applications due to

decreased power consumption, less parasitic capacitance, and enhanced switching speed. The total number of transistors would rise dramatically with the usage of an 8T or 10T full adder, further affecting area and power efficiency. Furthermore, in addition to requiring more silicon area, complete adders with greater transistor counts, such as 28T or 32T designs, would also increase leakage power and interconnect complexity, both of which are undesirable in contemporary circuit design. By utilizing transmission gate logic (TGL) and pass transistor logic (PTL) to optimize arithmetic and comparison operations while preserving a small footprint and improved performance, the suggested PTL-based method successfully addresses these issues. The circuit guarantees balanced drive strength and low latency by incorporating a meticulously tuned transistor scaling method, in which NMOS transistors are three times larger than their PMOS counterparts. The suggested comparator is a perfect choice for low-power, high-speed applications in contemporary VLSI design.

B. Power Dissipation

In contrast to conventional CMOS-based comparators, which use a greater number of transistors and suffer from higher switching power due to frequent charging and discharging of capacitances, the proposed design reduces the overall transistor count from 30 to 22, directly lowering dynamic power consumption by optimizing both dynamic and static power dissipation through an efficient pass transistor logic (PTL)-based approach. Additionally, the use of transmission gate logic (TGL) at the output minimizes short-circuit power by ensuring that PMOS and NMOS transistors do not conduct simultaneously for prolonged periods of time. The PTL-based construction lowers parasitic capacitances and superfluous switching activity, improving energy efficiency. However, complete adders with more transistors like those with 8T, 10T, or 28T designs introduce higher leakage currents and greater area needs, which raise static power dissipation. The comparison among different types of 1-bit comparators has been shown in TABLE I.

TABLE I. COMPARISON OF COMPARATOR TYPES

Comparator Type	Transistors
Shannon-based adder [2]	48
MCIT-7 T adder [2]	31
MUX-14T adder [2]	38
MUX-12T adder [2]	36
CMOS Logic [2]	30
Proposed	22

The power dissipation of many comparator systems with varied adder configurations is compared in the TABLE II. Using 12 and 14 transistors, respectively, the MUX-12T adder and MUX-14T adder are multiplexer-based complete adders. Utilizing seven transistors, the MCIT-7T adder offers a more effective substitute for bigger adder designs by minimizing power dissipation without sacrificing functionality. Furthermore, a visual chart of performance parameters for comparison of proposed circuit with other comparator types is shown in Figure 5. Which gives a clear

understanding of difference in power dissipation among the comparator types.

TABLE II. COMPATISON OF POWER DISSIPATION, PROPAGATION DELAY AND PDP OF THE PROPOSED COMPARATOR

Comparator Type	Power Dissipation (μW)	Propagation Delay (ps)	PDP (fJ)
Shannon-based adder [2]	250.6	19	4.761
MCIT-7 T adder [2]	51.66	16	0.826
MUX-14T adder [2]	98.5	17	1.674
MUX-12T adder [2]	51.33	7	0.359
CMOS Logic [2]	17.71	8	0.141
Proposed	7.3	5.5	0.04

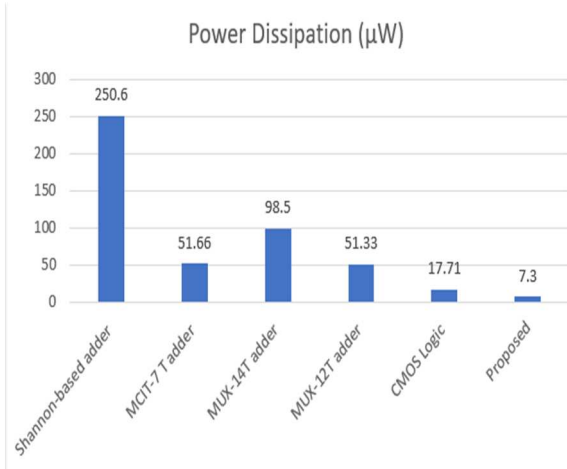


Fig. 5. Comparison of performance parameters.

Shannon-based adder uses Shannon's expansion theorem to optimize logic, but because of its intricate implementation, it has greater power dissipation (250.6 μW). When compared to other topologies, the suggested design (7.30 μW) performs better overall and exhibits higher power efficiency.

C. Propagation Delay

The time it takes for a signal to move from a logic gate's or circuit element's input to its output is known as the propagation delay in VLSI. It is a crucial performance indicator that has a direct impact on a digital system's overall speed. The type and quantity of transistors employed, the length of the link, the load capacitance, and the logic style's switching properties are some of the variables that affect propagation delay.

Depending on their transistor count and design complexity, comparators with various adder architectures have greatly variable propagation delays. The multiplexer-based structures of comparators that use MUX-12T and MUX-14T adders balance speed and power efficiency, resulting in low propagation delays. However, because of additional signal routing and switching costs, the delay may increase somewhat as the transistor count rises (for example, from 12T to 14T). However, as fewer transistors often lead to faster signal propagation and less capacitive loading, the MCIT-7 T adder, with its fewer transistors and better transmission gate design, is probably going to provide lower propagation delays.

D. Power- Delay Product (PDP)

A crucial parameter in VLSI design, the Power-Delay Product (PDP) measures the trade-off between digital circuit performance and power usage. It provides a measure of the design's energy efficiency and is computed as the product of the circuit's power dissipation and propagation delay. A reduced PDP is a crucial metric for assessing how well low-power design strategies work because it shows that the circuit can operate with less energy while still maintaining a respectable speed. Because PDP captures both a design's speed and energy consumption, it is very helpful when comparing various circuit layouts or technologies. PDP assists designers in finding a balance between maximizing speed and reducing power usage. A circuit with a high-power consumption and a very low propagation delay, for instance, might have a PDP comparable to a slower but more power-efficient design. In contemporary VLSI systems, where increased clock frequencies and decreasing transistor sizes frequently result in higher power densities and thermal difficulties, this trade-off is particularly crucial. Designers can create energy-efficient solutions that satisfy the performance needs of applications ranging from embedded devices to high-performance computing by concentrating on lowering the PDP.

E. Comparator Waveform

The circuit's reaction to various input combinations of A and B is shown by the waveform of the 1-bit comparator running at a supply voltage of 1.8V in Figure 6. The output values, which show how well the comparator can differentiate between the input signals, are shown by the Y-axis. The waveform indicates whether A is bigger than, equal to, or less than B by displaying the output's transitions as the inputs vary. The comparator exhibits effective functioning at 1.8V with distinct and steady output transitions, underscoring its dependability at low voltage levels. This waveform is crucial for assessing the comparator's performance, especially in low-power applications where it's crucial to retain speed and accuracy at lower supply voltages.

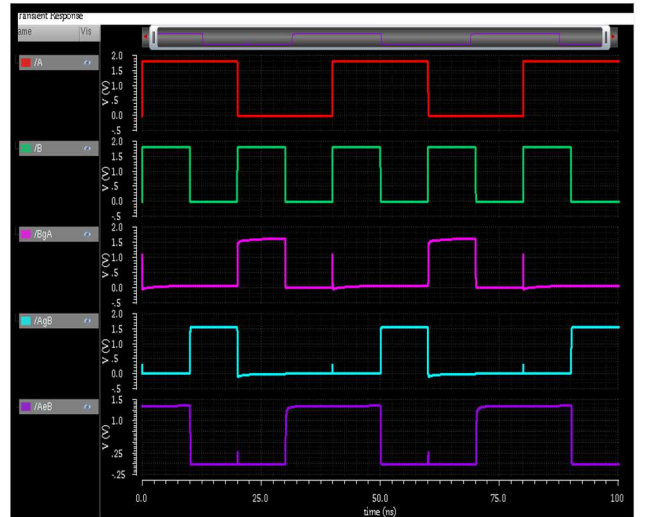


Fig. 6. Waveform of proposed 1-bit Comparator. The pink waveform represents the condition $A < B$, the blue waveform indicates $A > B$ and the purple waveform shows the condition $A = B$. The inputs A and B are the top two waveforms.

V. CONCLUSION

In conclusion, a low-power and effective digital circuit design is successfully demonstrated by the development of a

1-bit comparator employing a MUX-6T full adder with Pass Transistor Logic and Gate-Diffusion Input approaches, together with Transmission Gate Logic for the AND gates. The work demonstrates how combining different logic models can improve performance, minimize leakage current, and optimize power usage. As observed from TABLE I and TABLE II, the power dissipation shows a significant reduction of 58.7% from 17.71 μ W to 7.3 μ W and the propagation delay is reduced by 31.2% from 8ps to 5.5ps. The comparator is appropriate for energy-constrained applications like mobile communications and Internet of Things devices since it operates with stable and dependable performance at a supply voltage of 1.8V. This effort opens the door for further developments in energy-efficient computing by highlighting the significance of creative circuit design in producing high-performance, low-power digital computers. Its capacity to precisely differentiate between input signals while consuming the least amount of power is further validated by the successful simulation and waveform analysis of the comparator.

A one-bit comparator utilizing Transmission Gate Logic (TGL) and a MUX-6T design is shown in this study. An effective trade-off between transistor counts and switching activity results in lower power dissipation and propagation latency. Although it works well for single-bit comparison, the design might not scale well to multi-bit systems because to signal loss and increasing complexity, necessitating additional architectural or hybrid logic optimization.

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